Interpreting ML Results

A Manager's Exam Playbook — every question type, and the thinking behind each answer

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WHAT THIS GUIDE IS (AND ISN'T)

Ma'am's brief for the exam: **no coding — interpret results and take decisions.** You will be shown outputs (a probability equation, a confusion matrix, two models' scores, a tree, an AUC) and asked: what does this mean, and what should the business do?

So this guide is organised by **question type**, not by topic. For each type you get: how to recognise it, the exact thinking steps (narrated, so you learn the *method*, not just the answer), a fully worked example with numbers, a model answer you can reproduce in the exam, and the trap examiners set. It builds directly on your three earlier guides — nothing here contradicts them; this is the exam-day layer on top.

The nine question types

#	Question type	The manager skill being tested
1	“What is the probability?”	Turn a logistic equation into a % chance, then a decision (e.g. bad-loan probability)
2	“Interpret this coefficient”	Read sign, size (odds ratio) and business meaning of a model's numbers
3	“Build the confusion matrix”	Convert a word-problem into TP/FP/FN/TN, then compute accuracy, precision, recall/sensitivity, specificity, F1
4	“Which metric matters here?”	Match the business context to the right metric via the cost of each error
5	“Two models, two results — choose”	Compare models, spot the accuracy paradox, decide and justify
6	“Should we move the threshold?”	Tune the cut-off using error costs (with a rupee calculation)
7	“Read the decision tree”	Walk a case down a tree, pick the best split, spot overfitting
8	“Interpret the ROC / AUC”	Grade and compare models with one number; know when it lies
9	“Read the output table”	p-values, R^2 vs RMSE, AIC — say what's significant and what's good

Part 0 — The 5-step method behind every answer

Every interpretation question, whatever its costume, is answered by the same five moves. Learn this once and every section that follows is just the method applied to a different output.

Step	Ask yourself	What it looks like in an answer
1. Frame	What is the **positive class**? What decision hangs on this?	“Here positive = 'customer will default'. The decision is approve vs reject.”
2. Locate	Which numbers on the page actually answer the question?	Pick out the coefficient / the matrix cell / the recall value. Ignore decoration.
3. Translate	Say each number as a plain-English sentence.	“Recall 80% = of every 10 real defaulters, we catch 8 and miss 2.”
4. Cost it	Which error hurts this business more — the false alarm (FP) or the miss (FN)?	“A missed defaulter costs the loan amount; a false alarm costs one good customer's interest.”
5. Decide	State the action + one-line justification tied to the cost.	“Deploy Model B: its higher recall protects us from the costlier error.”

WHY STEP 4 IS THE HEART OF EVERY MARKS SCHEME

Steps 1–3 show you can read the output. Steps 4–5 show you can **manage** with it — and that is what an MBA course grades. Almost every 'take a decision' question is secretly asking one thing: **which mistake is more expensive here, the false alarm or the miss?** Answer that, and the metric to use, the model to pick, and the threshold to set all follow automatically.

EXAM HOOK — THE TWO SENTENCES THAT SCORE EVERYWHERE

“Precision protects us from false alarms; recall protects us from misses.”
“We choose the metric that guards against the costlier error for this business.” Write these (adapted) in nearly any evaluation answer and you have shown the examiner the core logic.

Question Type 1 — “What is the probability?”

How to recognise it: you are given a fitted logistic equation (log-odds = b0 + b1x1 + ...) and a specific person/case, and asked for the chance of an event — “what is the probability this customer's loan goes bad?” Sometimes a threshold is given and you must also make the call.

The recipe (memorise as three steps)

Step 1 — Raw score: plug the case's values into the straight line to get the log-odds z. **Step 2 — Squash:** convert z to a probability with the sigmoid, $p = e^z \div (1 + e^z)$. **Step 3 — Decide:** compare p to the threshold (default 0.5 unless told otherwise) and say Yes/No.

MY THINKING, NARRATED — WORKED EXAMPLE: BAD-LOAN PROBABILITY

The question: A bank's default model (positive = will default) is: log-odds = −3.2 + 0.05 × (Debt-to-income %) + 0.8 × (PastDefault, 1/0). A customer has DTI = 40% and one past default. The bank rejects any applicant whose default probability exceeds 30%. What is the probability, and what is the decision?

How I read it before touching a calculator: (Frame) positive = default, decision = approve/reject, and note the threshold is 0.30, not 0.50 — the bank is cautious, exactly as Section 6 of your logistic guide predicted. (Locate) I need three numbers from the equation: −3.2, 0.05, 0.8 — and two from the customer: 40 and 1.

Step 1 — raw score: z = −3.2 + 0.05×40 + 0.8×1 = −3.2 + 2.0 + 0.8 = **−0.4**. Sense-check before moving on: z is slightly negative, so p should land slightly below 50%. If my final answer is 0.85 I have made an arithmetic slip.

Step 2 — squash: $e^{-0.4} \approx 0.67$, so p = 0.67 ÷ (1 + 0.67) = 0.67 ÷ 1.67 ≈ **0.40**. Plain English: this customer has about a 40% chance of defaulting.

Step 3 — decide: 40% > the 30% cut-off → **reject (or refer for manual review)**. Notice the customer would have been APPROVED under the default 0.5 threshold — examiners love this twist, so always check which threshold the question actually gives you.

MODEL ANSWER (WHAT YOU WRITE IN THE EXAM)

z = −3.2 + 0.05(40) + 0.8(1) = −0.4. Probability of default = $e^z \div (1 + e^z)$ at z = −0.4 ≈ 0.40, i.e. a 40% chance this loan goes bad. Since 0.40 exceeds the bank's 0.30 threshold, the model classifies the applicant as a likely defaulter → reject/refer. (Under a default 0.5 threshold the same applicant would be approved — the business's chosen threshold, not the model, makes the final call.)

The sigmoid ready-reckoner (no calculator needed)

Exams reuse the same handful of z values. Memorise this row and you can squash almost anything:

z (raw score)	−3	−2	−1	−0.5	0	+0.5	+1	+2	+3
Probability p	0.05	0.12	0.27	0.38	0.50	0.62	0.73	0.88	0.95

EXAM HOOK — SYMMETRY SAVES YOU

σ(−z) = 1 − σ(z). Know σ(1) = 0.73 and you instantly know σ(−1) = 0.27. And z = 0 always gives exactly 0.50 — the tipping point. If a question's z lands between table values, interpolate roughly; interpretation questions reward the right ballpark + right decision, not six decimals.

TRAPS IN THIS QUESTION TYPE

• **Reporting the log-odds as the probability.** z = −0.4 is NOT '−4% chance' — probabilities can't be negative. Always squash.

• **Odds vs probability.** If asked for the *odds*, answer p÷(1−p): here 0.40÷0.60 = 0.67, i.e. roughly 2 defaults for every 3 repayments. '4-to-1 odds' means p = 80%, never p = 4.

• **Using 0.5 when the question set a different threshold.** Read the business rule; it outranks the default.

Question Type 2 — “Interpret this coefficient”

How to recognise it: you are shown a fitted model's coefficients and asked “what does the 0.8 on PastDefault mean?” or “interpret the negative sign on YearsEmployed.” No calculation of a case — just meaning.

The three things to say about any coefficient

Layer	What to check	How to phrase it
1. Sign	+ pushes the chance of the positive class up; − pushes it down	“PastDefault's coefficient is positive: a past default raises the chance of a new default, holding other factors constant.”
2. Size (odds ratio)	e raised to the coefficient = how the ODDS multiply per one-unit rise	$e^{0.8} \approx 2.2$ → “a past default roughly doubles the odds of default.” OR > 1 raises odds, < 1 lowers, = 1 no effect.
3. The S-curve caveat	The effect on the log-odds is constant, but the effect on the PROBABILITY is not	“The same one-unit change moves the probability a lot near p = 50% and barely at all near 0% or 99% (the 'swing' effect).”

MY THINKING, NARRATED — A NEGATIVE COEFFICIENT

The question: in the same default model, YearsEmployed has coefficient −0.5. Interpret it for the credit committee.

Layer 1 (sign): negative → more years employed means LOWER chance of default. Sensible — job stability. Always sanity-check the sign against intuition; if it fights intuition, say so, because examiners sometimes plant an odd sign to see if you notice.

Layer 2 (size): odds ratio = $e^{-0.5} \approx 0.61$. Plain English: each extra year of employment multiplies the default odds by 0.61 — i.e. cuts them by about 39%. Note I said **odds**, not probability. The coefficient adds to the log-odds; the odds ratio multiplies the odds; neither is a straight subtraction from the probability.

Layer 3 (caveat): for a borderline applicant (p near 50%) one more year of employment shifts the decision a lot; for a near-certain repayer (p near 5%) it changes almost nothing. Same push, different spot on the swing.

Dummy variables: if the variable is a 1/0 dummy (like PastDefault), interpret it as a group comparison: “defaulters-in-the-past have about 2.2× the odds of defaulting again compared with otherwise-identical applicants with no past default.”

MODEL ANSWER TEMPLATE

“The coefficient on [X] is [sign], so higher [X] [raises/lowers] the likelihood of [positive class], other factors held constant. Its odds ratio is e^[b] ≈ [value]: each one-unit increase multiplies the odds of [event] by [value] (≈ [%] change). The effect on the probability itself is largest for borderline cases near 50% and small near the extremes.” — three sentences, three layers, full marks.

EXAM HOOK

$e^{\text{coefficient}}$ = odds ratio. >1 helps the event, <1 hurts it, =1 no effect. And the classic true/false: “a logistic coefficient has a constant effect on the probability” — FALSE; constant on the log-odds only.

Question Type 3 — “Build the confusion matrix and compute the scores”

How to recognise it: a word-problem gives you totals in prose — “tested on 500 customers... flagged 120... 80 of those actually churned” — and asks for accuracy, precision, recall/sensitivity, specificity or F1. The whole battle is translating words into the four cells; the formulas are then one line each.

The translation dictionary (words → cells)

Phrase in the question	What it is	Cells involved
“Actually churned / truly had the disease”	Actual positives (row total)	TP + FN
“Model flagged / predicted / raised an alert”	Predicted positives (column total)	TP + FP
“Correctly identified / caught”	True positives	TP
“Missed / slipped through / wrongly cleared”	False negatives (Type II)	FN
“False alarm / wrongly flagged / needless worry”	False positives (Type I)	FP
“Correctly cleared / rightly ignored”	True negatives	TN

MY THINKING, NARRATED — WORKED EXAMPLE: CHURN MODEL

The question: A telecom churn model is tested on 500 customers. In reality 100 churned. The model flagged 120 customers as churners, and 80 of those flagged really did churn. Compute accuracy, precision, recall (sensitivity), specificity and F1, and interpret each for the retention head.

How I fill the matrix — always in this order: (1) TP first: “80 of the flagged really churned” → TP = 80. (2) FP from the flagged total: flagged 120, of which 80 correct → FP = 120 − 80 = 40. (3) FN from the actual total: 100 real churners, 80 caught → FN = 100 − 80 = 20. (4) TN is whatever remains: 500 − 80 − 40 − 20 = 360. Cross-check: rows and columns must re-add to the question's totals (80+20 = 100 ✓, 80+40 = 120 ✓).

Now the scores, each with its plain-English sentence (this is where interpretation marks live):

• Accuracy = (80+360)÷500 = **88%** — “we got 88 of every 100 customers right overall.”

• Precision = 80÷120 = **66.7%** — “when we flag someone as a churner, we're right 2 times in 3; 1 in 3 retention offers goes to someone who was staying anyway.”

• Recall / Sensitivity = 80÷100 = **80%** — “of the customers genuinely about to leave, we catch 8 in 10 and miss 2.” (Sensitivity IS recall — same number, medical name.)

• Specificity = 360÷(360+40) = **90%** — “of the loyal customers, we correctly leave 9 in 10 alone.”

• F1 = 2×(0.667×0.80)÷(0.667+0.80) = 1.067÷1.467 ≈ **0.73** — “the balanced single grade; respectable, because neither precision nor recall is weak.”

MODEL ANSWER — THE INTERPRETATION PARAGRAPH THAT FOLLOWS THE ARITHMETIC

“The model is a good catcher (recall 80%) but a moderately wasteful flagger (precision 67%). For churn, a miss (losing a customer we could have saved) usually costs more than a false alarm (a discount to a loyal customer), so the recall of 80% is the headline strength. If the retention budget is tight, we would instead work on precision — or raise the threshold so we only call the most certain churners.”

TRAPS

• **Swapping Type I and Type II.** FP = Type I (false alarm, 'cried wolf once'); FN = Type II (the real one slipped past).

• **Dividing precision by the actual total or recall by the flagged total.** Precision's denominator is what the MODEL said (TP+FP); recall's is what REALITY says (TP+FN).

• **Forgetting the cross-check.** Ten seconds of re-adding rows/columns catches most exam-hall slips.

Question Type 4 — “Which metric matters in THIS business?”

How to recognise it: a scenario, then “which metric should the manager prioritise, and why?” There is exactly one decision rule. Everything else is applying it.

> **THE ONLY RULE YOU NEED**
> Ask: **what does a false alarm (FP) cost, and what does a miss (FN) cost?** If the MISS is costlier → prioritise **RECALL** (catch them all, tolerate false alarms). If the FALSE ALARM is costlier → prioritise **PRECISION** (only cry wolf when sure, tolerate some misses). If both matter and the data is imbalanced → report **F1**. Only when classes are balanced AND errors cost about the same is plain **ACCURACY** a safe headline.

The scenario bank (learn the justifications, not just the answers)

Business scenario (positive class)	Costlier error	Metric	One-line justification
Cancer / disease screening (has disease)	FN — a missed patient may die	Recall	A false alarm is just a re-test; a miss can be fatal.
Loan default screening (will default)	FN — an approved defaulter loses the principal	Recall	A missed defaulter costs the loan; a false alarm costs one good customer's interest.
Fraud detection (transaction is fraud)	FN — fraud goes through	Recall (then precision)	Missed fraud is direct loss; but too many false blocks anger customers, so precision is the secondary check.
Spam filter (mail is spam)	FP — a real email is binned	Precision	Losing a genuine offer letter is far worse than seeing one extra spam.
Courtroom conviction (guilty)	FP — an innocent person jailed	Precision	The system is built to avoid false positives at almost any cost.
Marketing campaign with tight budget (will respond)	FP — money spent on non-responders	Precision	Every wasted mailer burns budget; missing a would-be buyer is cheaper.
Retention offers, high customer LTV (will churn)	FN — a savable customer walks	Recall	The discount (FP cost) is small next to lifetime value lost (FN cost).
Airport security screening (is a threat)	FN — catastrophic	Recall	Massive tolerance for false alarms; zero tolerance for misses.
Factory quality check, cheap rework (defective)	FN — defect reaches customer	Recall	Re-inspecting a good unit is pennies; a shipped defect costs warranty + reputation.
Shortlisting from thousands of CVs (fit for role)	FP — interviews waste panel time	Precision	You can't interview everyone; each slot must count. (Note misses = lost talent, so state the trade-off.)

MY THINKING — WHY 'LOAN DEFAULT' CAN GO EITHER WAY (THE NUANCE THAT IMPRESSES)

The answer depends on how the positive class is framed and on margins. Framed as 'predict default' (above), the miss is the costly error → recall. But if the bank's margins are thin and it competes on approval rates, rejecting good customers (FPs) has a real cost too — so a full answer says: “prioritise recall on defaults, but monitor precision, and set the final balance with a cost-based threshold (Type 6).” Naming BOTH costs before choosing is exactly the 'manager, not calculator' behaviour the course rewards.

EXAM HOOK — WHERE IS ACCURACY THE RIGHT ANSWER?

When classes are roughly balanced and both errors cost about the same — e.g. classifying products into two equally common categories for routing. The moment the question says 'rare', 'only 2% of cases', or implies asymmetric costs, accuracy is the WRONG headline (see Type 5's paradox).

Question Type 5 — “Two models, two results — which do you deploy?”

How to recognise it: a table with Model A and Model B and 2–5 metrics each, plus a business setting. Ma'am flagged this one explicitly. The trick: the models will usually disagree — one wins on accuracy, the other on recall — and the setting tells you which disagreement matters.

The 4-step comparison routine

(1) Identify the positive class and the costlier error (Type 4's rule). (2) Compare the metric that guards that error. (3) If the 'worse' model has higher accuracy, name the **accuracy paradox** and explain it in one line. (4) Decide, justify by cost, and offer the threshold lever as the refinement.

MY THINKING, NARRATED — CASE A: THE ACCURACY PARADOX IN A FRAUD SETTING

The question: Only 4% of transactions are fraudulent. Two models are tested: Model X — Accuracy 96%, Precision 90%, Recall 20%. Model Y — Accuracy 90%, Precision 55%, Recall 85%. Which should the bank deploy?

Step 1 (frame + cost): positive = fraud; the miss (FN) lets real fraud through — direct monetary loss. Costlier error = the miss → the metric that matters most is **recall**.

Step 2 (compare on recall): X catches only 20% of fraud — it misses 4 of every 5 frauds. Y catches 85%. On the metric that matters, Y wins by a mile.

Step 3 (name the paradox): X's 96% accuracy is a mirage. With fraud at 4%, a model that says 'never fraud' scores 96% accuracy while catching nothing — X is barely better than that lazy model. High accuracy on rare-event data mostly measures how common the majority class is, not skill. This is the **accuracy paradox**, the single most-tested idea in this course.

Step 4 (decide + refine): Deploy Y. Its lower precision (55%) means ~45% of its alerts are false alarms — acceptable, because an alert triggers a cheap review call, while a miss loses the transaction amount. If the review team drowns in alerts, raise Y's threshold slightly to trade a little recall for precision.

MODEL ANSWER (CASE A, EXAM-READY)

“With fraud at 4%, accuracy is misleading (a model predicting 'never fraud' would score 96% — the accuracy paradox). The costly error here is the miss, so recall is the deciding metric: Y catches 85% of fraud vs X's 20%. Deploy Model Y; its false alarms cost only review effort, while X's misses cost the fraud amounts themselves. Fine-tune Y's threshold if review capacity becomes the bottleneck.”

CASE B — SAME ROUTINE, OPPOSITE ANSWER (BUDGET MARKETING)

Two lead-scoring models for a costly direct-mail campaign: P has Precision 80% / Recall 40%; Q has Precision 45% / Recall 85%. Costlier error now = the FALSE ALARM (each mailer to a non-buyer burns budget), so precision decides → choose P: 8 of every 10 mailers reach genuine prospects. Identical routine as Case A, flipped by the cost structure — proof that you're applying a method, not memorising 'recall is always better'.

CASE C — WHEN THE TABLE GIVES AUCs INSTEAD

Model A AUC 0.82, Model B AUC 0.68 → “A is the better ranker across ALL thresholds — it separates the classes more cleanly regardless of where we later cut.” Bands: ≈0.5 useless, 0.6–0.7 mediocre, 0.7–0.85 very good, 0.85+ excellent, 0.999 suspicious (data leakage). One caveat earns a bonus mark: on extremely rare positives, AUC can still flatter — check precision/recall/F1 too.

TRAPS

• Picking the higher-accuracy model 'because accuracy is the overall measure' — on imbalanced data this is exactly the planted wrong answer.

• Comparing precision of one model with recall of the other. Compare like with like, on the metric the business cares about.

• Forgetting F1 as the tie-breaker when the question insists both errors matter: F1 punishes whichever of precision/recall is weaker, so a model can't hide a terrible recall behind a shiny precision.

Question Type 6 — “Should we move the threshold?”

How to recognise it: the model is fixed; the question is where to cut. “The hospital wants to miss fewer cases — what should it do to the 0.5 cut-off, and what is the trade?” Or a small cost table asks which threshold minimises total cost.

The two direction rules (never fail these)

Move	What happens	When a business does it
LOWER the threshold (say 0.5 → 0.3)	Model says 'Yes' more easily → Recall ↑, FN ↓, but FP ↑, Precision ↓	When the miss is deadly/expensive: disease screening, fraud, safety checks
RAISE the threshold (say 0.5 → 0.8)	Model says 'Yes' only when very sure → Precision ↑, FP ↓, but FN ↑, Recall ↓	When the false alarm is expensive: big-ticket loan approvals, costly interventions, convictions

MY THINKING, NARRATED — THE RUPEE CALCULATION

The question: In a screening programme, a missed case (FN) costs ₹100 (late treatment) and a false alarm (FP) costs ₹5 (one extra test). At threshold 0.5 the model produces FN = 30, FP = 10. At threshold 0.3 it produces FN = 10, FP = 60. Which threshold should the programme use?
Locate the machinery: total cost = (cost per FP × number of FPs) + (cost per FN × number of FNs). Nothing else in the story matters.
Threshold 0.5: 10×₹5 + 30×₹100 = ₹50 + ₹3,000 = **₹3,050**.
Threshold 0.3: 60×₹5 + 10×₹100 = ₹300 + ₹1,000 = **₹1,300**.
Decide: 0.3 wins — total error cost falls by ₹1,750 even though we now raise six times as many false alarms. That is the whole logic of cost-based thresholds: because a miss is 20× as expensive as a false alarm, we deliberately buy lots of cheap FPs to avoid a few ruinous FNs.
The sanity narrative to close with: lowering the cut-off moved us along the ROC curve toward higher sensitivity — the direction the cost structure demanded.

MODEL ANSWER PHRASES

“Lower the threshold: sensitivity/recall rises and misses fall, at the accepted price of more false alarms and lower precision.”
“Choose the threshold minimising C1×FP + C2×FN; here 0.3 gives ₹1,300 vs ₹3,050 at 0.5, so 0.3.”
If the question wants a 'balanced' cut instead of a costed one: “pick the threshold maximising Youden's J = Sensitivity + Specificity − 1” (e.g. 0.85 + 0.70 − 1 = 0.55) — the ROC point furthest above the diagonal.

EXAM HOOK

The threshold is a BUSINESS dial, not a statistical constant. 0.5 is only the default. Screening lowers it; high-stakes approvals raise it. And moving the threshold changes the confusion matrix — it does NOT retrain or improve the model itself (its AUC is unchanged).

Question Type 7 — “Read the decision tree and decide”

How to recognise it: a drawn tree (or its rules) plus a new case to classify; or candidate splits with gains/impurities and “which becomes the root?”; or train-vs-test accuracies and “what's wrong?”

Sub-type 7a — Walk a case down the tree

MY THINKING, NARRATED

The tree (a loan model): Root: Is DTI > 45%? — Yes → leaf 'Default' (leaf purity: 70% of training cases here defaulted). No → next question: Past default? — Yes → leaf 'Default' (55%). No → leaf 'Repay' (90% repaid).
The case: applicant with DTI 38%, one past default.
The walk — one question at a time, top down: DTI 38% ≤ 45% → take the No branch. Past default = Yes → leaf 'Default (55%)'. **Prediction: default, with probability ≈ 0.55.**
The manager's sentence (this is the interpretation mark): “The tree is an IF–THEN policy anyone can audit: IF DTI ≤ 45% AND there is a past default THEN treat as risky (55%). At only 55% purity this leaf is barely better than a coin flip — so route these applicants to manual review rather than auto-reject.” Reading the leaf's purity as a CONFIDENCE level, not just a label, is what separates a manager's answer from a mechanical one.

Sub-type 7b — Which attribute becomes the root?

MY THINKING, NARRATED

Given: Information Gain of Income = 0.12, DTI = 0.38, Age = 0.05. The tree greedily picks the attribute with the **highest gain** (equivalently the lowest weighted child impurity) → **DTI becomes the root**. Plain English for the examiner: gain measures how much messiness (entropy) the question removes; DTI's question sorts defaulters from repayers most cleanly in one cut, like the best first question in Twenty Questions. If the numbers given are Gini impurities instead, pick the LOWEST — minimum impurity and maximum gain always point to the same winner (your Decision Tree guide's Patrons-vs-Type example: gain 0.28 beats 0).

Sub-type 7c — Spot the overfit

MY THINKING, NARRATED

Given: a 14-level tree scores 99% on training data but 68% on test data. Diagnosis in one line: the huge train–test gap = **high variance = overfitting** — the tree memorised noise instead of learning the pattern (the student who memorised last year's answers). Prescriptions, in exam order: (1) prune — pre-pruning caps growth via maxdepth / minsplit (min cases a parent needs to split) / minbucket (min cases in a leaf), or post-pruning grows fully then trims branches that don't help a validation set; (2) or move to a **random forest** — many trees on bootstrap samples with random feature subsets, voting — which averages the quirks away.

EXAM HOOK — TREE ONE-LINERS

• Root = first split, sees all data, exactly one. Leaf = terminal, gives the prediction.
• Highest information gain = lowest weighted impurity = best split.
• Big train–test accuracy gap = overfitting = high variance → prune or ensemble.
• Trees only make axis-parallel (one-variable, straight) cuts — their headline weakness; ensembles/SVM handle curves.
• Random Forest = bagging + random feature subset (m = √K for classification); trees vote in parallel. Boosting builds sequentially, each tree correcting the last.

Question Type 8 — “Interpret the ROC curve / AUC”

How to recognise it: an ROC plot, or just AUC numbers, and “comment on the model” / “which is better?”

What the picture is

The ROC curve tries EVERY threshold at once and plots, for each: good catches (True Positive Rate = recall = sensitivity) going UP, against false alarms (False Positive Rate = 1 − specificity) going RIGHT. Up = good, right = bad, so the ideal curve hugs the top-left corner. The diagonal is a coin-flip. AUC compresses the whole curve into one grade — and it has a lovely plain-English meaning: **the probability the model ranks a random real-Yes above a random real-No.**

AUC	Verdict	What you say as a manager
≈ 0.5	Useless	“No better than a coin flip — the model has learned nothing; go back to the features.”
0.6 – 0.7	Mediocre	“Some signal, not deployable alone; use as one input among several.”
0.7 – 0.85	Very good	“Deployable; now choose the operating threshold to match our error costs.”
0.85+	Excellent	“Strong separator — verify it holds on fresh data before trusting it.”
≈ 0.999	Too good to be true	“Red flag — almost certainly data leakage (the model peeked at the answer); audit the features.”
< 0.5	Worse than random	“Its ranking is inverted — flipping every prediction would beat it; something is wired backwards.”

MY THINKING — THE TWO SENTENCES THAT COVER ANY AUC QUESTION

First: state the grade from the band table and its plain meaning ('AUC 0.82 = very good: given one true defaulter and one true repayer, the model scores the defaulter higher 82% of the time'). Second: state AUC's superpower and its blind spot — it is **threshold-independent** (compares models before any cut-off is chosen), but on extremely rare positives it can still flatter, so confirm with precision/recall/F1. Grade → meaning → caveat: full marks.

Question Type 9 — “Read this output table”

How to recognise it: a software regression/logistic summary — Estimates, p-values, stars, R^2, RMSE, F-statistic, AIC — and “which variables matter? Is the model good?” Scan in this exact order:

Look at	The rule	The sentence you write
p-value of each variable (t/Wald test)	< 0.05 → significant (stars help: * ** ***)	“Experience (p = 0.003) genuinely affects salary; Age (p = 0.41) shows no reliable effect — we cannot rule out zero.”
F-test (linear) / LR test (logistic)	< 0.05 → the model as a whole beats guessing	“The overall model is significant — at least one input is doing real work.”
R^2 vs Adjusted R^2	Adjusted penalises junk variables; use it to compare models	“R^2 never falls when variables are added, so we compare on Adjusted R^2.”
R^2 WITH RMSE, never alone	R^2 = does the model explain variation; RMSE = typical error in y's own units	“RMSE 5.9 with R^2 0.00 = just guessing the average; R^2 0.98 with RMSE 121 = explains pattern but predicts poorly. Judge on both.”

Look at	The rule	The sentence you write
AIC / BIC (comparing two models)	Lower = better; they punish needless complexity	“Model 2's lower AIC means better fit per unit of complexity — prefer it.”
VIF (if shown)	> 5 concern, > 10 must fix	“VIF 12 on Age and Experience → multicollinearity; drop one or combine (PCA).”

EXAM HOOK

t/Wald test = one variable; F/LR test = the whole model; 0.05 is the magic line. And the guaranteed pairing: never judge on R^2 or RMSE alone — the course's two planted counter-examples ($R^2 = 0$ with small RMSE; $R^2 = 0.979$ with huge RMSE) are both BAD models.

Rapid-fire drill — 10 one-minute questions (answers overleaf logic, right here)

Cover the right column and attempt the left. Every answer uses the 5-step method.

Question	Answer + the thinking in one breath
1. z = +1 for a customer. Probability of Yes?	σ(1) ≈ 0.73 → 73%. (Positive z → above 50%; z = 0 would be exactly 50%.)
2. Odds of default are 3-to-1. Probability?	p = 3÷(3+1) = 75%. Odds are Yeses per No, never a probability.
3. Coefficient on Income = −0.9. Odds ratio & meaning?	e^−0.9 ≈ 0.41: each extra unit of income multiplies default odds by 0.41, cutting them ~59%.
4. 1,000 emails, 20 spam. Model flags 25; 15 are truly spam. Precision? Recall?	TP 15, FP 10, FN 5 → Precision 15/25 = 60%; Recall 15/20 = 75%.
5. Same model: accuracy?	TN = 1000−15−10−5 = 970 → (15+970)/1000 = 98.5% — and it's flattering: 'never spam' scores 98%. Accuracy paradox.
6. Cancer test: which error is Type II and why is it feared?	FN — saying healthy when disease exists. Feared because the patient goes untreated.
7. Model A: acc 94%, recall 15%. Model B: acc 88%, recall 82%. Rare-disease screening — deploy which?	B. Miss is the costly error → recall decides; A's accuracy is the majority-class mirage.
8. Screening programme wants fewer misses. Threshold action + the price?	Lower it. Recall/sensitivity ↑, FN ↓; price = more false alarms, precision ↓.
9. Tree: train 98%, test 66%. Diagnosis + two fixes?	Overfitting (high variance). Prune (maxdepth/minsplit/minbucket or post-prune) or use a random forest.
10. AUC = 0.999 on loan data. Celebrate?	No — too good to be true; suspect data leakage (a feature that already contains the answer). Audit features.

The last-minute half-page

IF YOU REMEMBER ONLY THIS

• **Probability question:** line → z → sigmoid → p → compare to the QUESTION'S threshold. σ(0)=0.50, σ(1)=0.73, σ(2)=0.88; σ(−z)=1−σ(z).
• **Coefficient:** sign → direction; e^b → odds ratio (>1 raises odds); effect on probability biggest near 50% (the swing).
• **Matrix:** TP first, FP from flagged total, FN from actual total, TN = the rest; then cross-check. Precision = TP/(TP+FP) (model's claims); Recall = Sensitivity = TP/(TP+FN) (reality's cases); Specificity = TN/(TN+FP); F1 = harmonic blend that punishes the weaker one.
• **Metric choice:** costlier miss → recall (cancer, fraud, default, safety); costlier false alarm → precision (spam, conviction, tight budgets); imbalanced + both matter → F1; accuracy only when balanced & symmetric.
• **Two models:** frame → costlier error → compare THAT metric → name the accuracy paradox if planted → decide → offer threshold tuning.
• **Threshold:** lower = catch more, more false alarms; raise = surer Yeses, more misses. Costed: minimise C1×FP + C2×FN. Balanced: max Youden's J = Sens + Spec − 1.
• **Tree:** walk top-down one question at a time; leaf purity = confidence; highest gain/lowest impurity wins the split; train–test gap = overfit → prune or forest.
• **AUC:** 0.5 coin-flip, 0.7–0.85 very good, 0.999 leakage; threshold-independent; on very rare positives, double-check with F1.
• **Output table:** $p < 0.05$ significant; t/Wald = one variable, F/LR = whole model; compare on Adjusted R^2 and AIC; judge fit on R^2 AND RMSE together.
• **Every answer ends the same way:** a decision, justified by which error costs this business more.

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